Tudor Tomulescu

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Education

Masters in Modern Signal Processing Techniques (TMPS) *Oct 2025 - present*
Faculty of Electronics, Telecommunications and Information Technology (ETTI)

Microelectronics, Optoelectronics, and Nanotechnologies (MON) *Oct 2021 – July 2025*
Faculty of Electronics, Telecommunications and Information Technology (ETTI)

Technologies

Languages: C++, C, Python, Verilog, VHDL

Skills:

- Object-Oriented Programming
- Physical design & implementation —Layout Design
- Programming & HDLs
- Computer systems & embedded design — microcontroller architecture, embedded
- Orcad systems
- Integrated analog circuits

Projects

Software Application with Graphical User Interface for STM32 Discovery Systems using TouchGFX

- **Objective:** Developed a graphical user interface (GUI) for a distributed embedded system on a CAN bus, enabling real-time monitoring of network node status and system time (date/hour).
- **Roles & Achievements:** Managed full-stack HW/SW design, including TouchGFX Designer development with code generation for STM32CubeIDE; integrated FreeRTOS and implemented CANopen protocols (TIMESTAMP, HEARTBEAT); performed functional testing and final demonstrations.
- **Hardware Platform:** STM32F769-DISCO and STM32F429-DISCO (TFT 800×480, touchscreen).
- **Key Results:** Delivered a functional interface for coordinator and execution nodes (allowing HB parameter modification and status visualization), a network time synchronization application, and a real-time monitoring demo. Technologies: TouchGFX, STM32CubeMX/CubeIDE, FreeRTOS, CAN/CANopen, Quad-SPI/FMC.

MIPS Processor RTL Description and Verification

- **Objective:** Designed and verified a MIPS processor architecture at the Register Transfer Level (RTL) to ensure accurate instruction execution and data path routing.
- **Roles & Achievements:** Managed the hardware description and implementation using Verilog; developed a comprehensive functional verification environment to validate the processor's structural integrity and logic flow.
- **Tools & Technologies:** Verilog, SystemVerilog, QuestaSim, ModelSim, Xilinx Vivado.
- **Key Results:** Delivered a fully functional and verified RTL model of a MIPS processor, demonstrating strong proficiency in hardware description languages (HDLs) and industry-standard functional verification workflows.

Languages

- **Romanian:** native
- **English:** fluent